

Intro to Data-Driven Dynamics

Aim: Use data to make qualitative and/or quantitative predictions about dynamical systems - systems evolving in time.

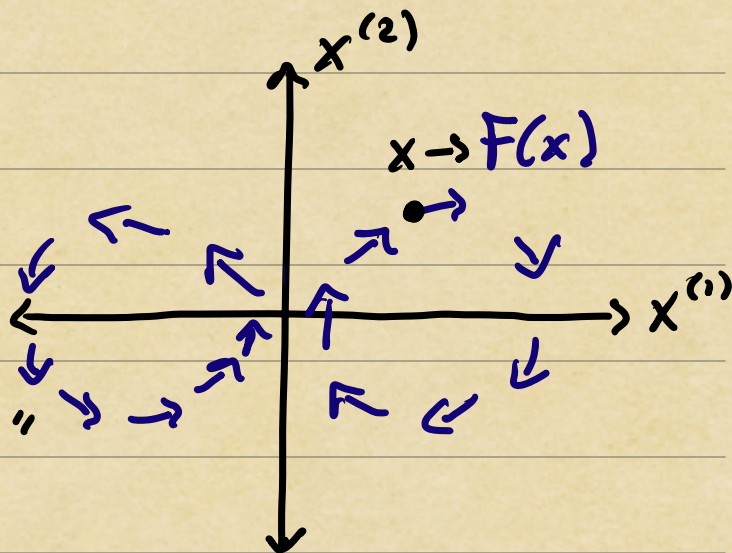
Autonomous DS (Continuous).

Given an open connected set $\Omega \subset \mathbb{R}^d$, and a map $F: \Omega \rightarrow \Omega$, an autonomous continuous-time dynamical system has form

$$(*) \quad \frac{dx}{dt} = F(x), \quad x_0 \in \Omega.$$

$x(t; x_0)$ - State of DS at time $t \geq 0$.

We often interpret F as a vector field on Ω . At point $x \in \Omega$, $F(x)$ describes the "velocity" of the state $x(t)$.



Given an initial state $x_0 \in \Omega$, the evolution of the state is called a trajectory:

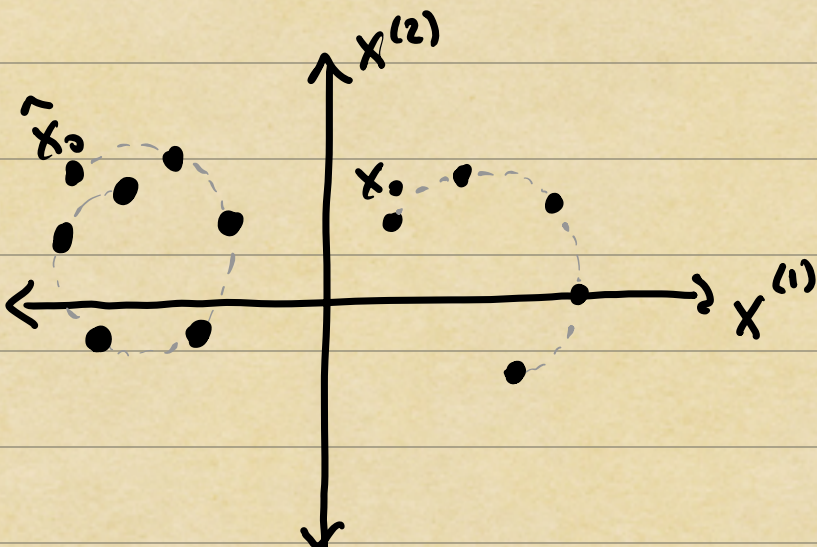
$$t \rightarrow X(t; x_0), \quad \text{for } t \geq 0.$$

Trajectory : Snapshot Data.

In the "data-driven" setting, we may not know much (or anything) about the map F governing the dynamical system.

Instead, we observe discrete samples from one or more trajectories, e.g.,

$$X_0 = X(0; x_0), X_1 = X(t_1; x_0), \dots, X_m = X(t_m; x_0).$$



Data sampled from two trajectories with initial states x_0 and $\hat{x}_0$

If $0 = t_0 < t_1 < \dots < t_m$ are equispaced with $\Delta t = t_{j+1} - t_j$, we can interpret the trajectory data as samples from a **discrete dynamical system**,

$$X_{j+1} = F^{\Delta t}(X_j), \quad X_0 \in \Omega.$$

In this case, it is useful to organize the data into a collection of input-output pairs for the map $F^{\Delta t}$:

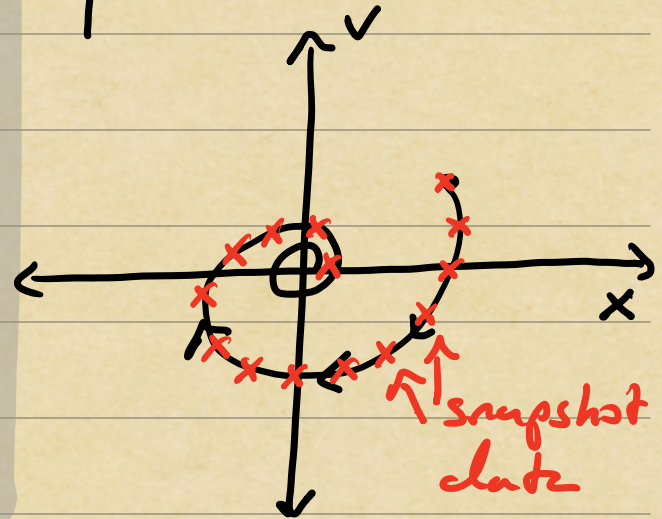
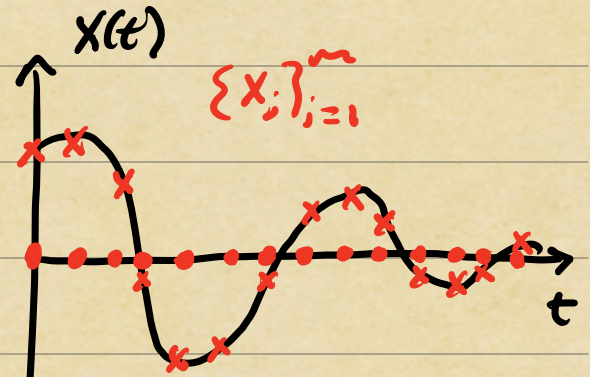
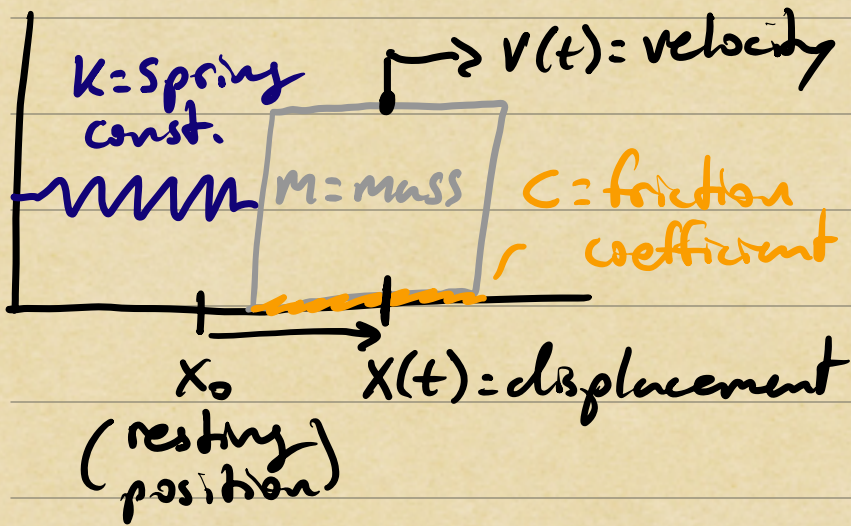
$$X = \begin{bmatrix} | & & | \\ x_1 & \dots & x_m \\ | & & | \end{bmatrix}, \quad Y = \begin{bmatrix} | & & | \\ y_1 & \dots & y_m \\ | & & | \end{bmatrix}$$

where $\underbrace{y_j = F^{\Delta t}(x_j)}_{\text{"snapshots"}}, \quad j=1, \dots, m.$

The "snapshots" can be formed from a single trajectory or multiple trajectories.

Example. Simple Oscillator

$$\frac{d}{dt} \begin{bmatrix} x \\ v \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -\frac{k}{m} & -\frac{c}{m} \end{bmatrix} \begin{bmatrix} x \\ v \end{bmatrix}$$



"trajectory" of state

Q: What questions could we ask the "snapshot" data about the oscillator?

$\Rightarrow$ Predict Trajectory

$\Rightarrow$ Frequency of oscillation

$\Rightarrow$ Damping/Dissipation rate

$\Rightarrow$ Response to external forcing?

$\Rightarrow$ Hidden/Unresolved variables?

Linear Dynamical Systems

The simple oscillator is an example of a linear dynamical system. A general linear dynamical system has the form

$$(*) \quad \frac{d}{dt} \begin{bmatrix} x_1 \\ \vdots \\ x_d \end{bmatrix} = \begin{bmatrix} a_{11} & \dots & a_{1d} \\ \vdots & & \vdots \\ a_{d1} & \dots & a_{dd} \end{bmatrix} \begin{bmatrix} x_1 \\ \vdots \\ x_d \end{bmatrix}$$

$$\dot{x} = A x$$

The key to understanding linear dynamics is hidden in the eigenvalues and eigenvectors of the matrix A .

Suppose that $Av_j = \lambda_j v_j$, for $j=1, \dots, d$ and that $\{v_1, \dots, v_d\}$ are lin. indep.

$\Rightarrow$ Such A is diagonalizable because it is diagonal in the eigenvector basis.

$$A \underbrace{\begin{bmatrix} 1 \\ \vdots \\ \vdots \\ \vdots \\ 1 \end{bmatrix}}_V = \underbrace{\begin{bmatrix} 1 \\ \vdots \\ \vdots \\ \vdots \\ 1 \end{bmatrix}}_V \underbrace{\begin{bmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_d \end{bmatrix}}_\Lambda \Rightarrow V^{-1}AV = \underbrace{\Lambda}_{\substack{\uparrow \\ \text{diagonal}}}$$

We can solve (**) explicitly by changing to eigenvector coords when A is diagonalizable (otherwise, one needs to use the Jordan normal form).

$$\text{Let } \begin{bmatrix} x_1 \\ \vdots \\ x_d \end{bmatrix} = \begin{bmatrix} 1 & & \\ \vdots & \ddots & \\ \vdots & & \vdots \\ 1 & & \end{bmatrix} \begin{bmatrix} c_1 \\ \vdots \\ c_d \end{bmatrix}$$

$$x = Vc$$

$$\text{so that } \frac{d}{dt} \begin{bmatrix} x_1 \\ \vdots \\ x_d \end{bmatrix} = \begin{bmatrix} 1 & & \\ \vdots & \ddots & \\ \vdots & & \vdots \\ 1 & & \end{bmatrix} \frac{d}{dt} \begin{bmatrix} c_1 \\ \vdots \\ c_d \end{bmatrix}$$

$$\dot{x} = V\dot{c}$$

$$\text{Then, } \dot{x} = Ax \Leftrightarrow \dot{c} = V^{-1}AVc = \Lambda c$$

$$\text{decoupled ODEs} \Rightarrow \dot{c}_j = \lambda_j c_j \quad \text{for } j=1, \dots, d$$

So, the solution in the eigenbasis is

$$\begin{bmatrix} c_1 \\ \vdots \\ c_d \end{bmatrix} = \underbrace{\begin{bmatrix} e^{\lambda_1 t} & & \\ & \ddots & \\ & & e^{\lambda_d t} \end{bmatrix}}_{e^{-\Lambda t}} \begin{bmatrix} c_1(0) \\ \vdots \\ c_d(0) \end{bmatrix},$$

and transforming back via $x = Vc$,

$$x(t) = \underbrace{V e^{-\Lambda t} V^{-1}}_{e^{At}} x(0).$$

In particular, the discrete map for time increments $\Delta t > 0$ is given by

$$x(t + \Delta t) = e^{A \Delta t} x(t).$$

The matrix function $e^{At} = V e^{-\Lambda t} V^{-1}$ is called the matrix exponential.

Example. Harmonic Oscillator

Returning to the simple oscillator, let $k=m$ and $c=0$ (no friction), so that

$$\frac{d}{dt} \begin{bmatrix} x \\ v \end{bmatrix} = \underbrace{\begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}}_A \begin{bmatrix} x \\ v \end{bmatrix}.$$

The eigenvalues of A are the soln's of

$$\det(A - \lambda I) = \lambda^2 + 1 = 0.$$

$$\Rightarrow \lambda_{\pm} = \pm i$$

The eigenvectors are null vectors:

$$(A \mp iI) v_{\pm} = 0$$

$$\Rightarrow v_{\pm} = \begin{bmatrix} 1 \\ \pm i \end{bmatrix}$$

With initial condition $x(0)=1, v(0)=0$,

$$\begin{bmatrix} x(t) \\ v(t) \end{bmatrix} = \frac{e^{it}}{2} \begin{bmatrix} 1 \\ i \end{bmatrix} + \frac{e^{-it}}{2} \begin{bmatrix} 1 \\ -i \end{bmatrix}$$

$$= \begin{bmatrix} \cos t \\ \sin t \end{bmatrix}$$

Q: How do the eigenvalues, and trajectories, change if $c > 0$?